



HIGHER DIPLOMA IN COMPUTER SCIENCE (NFQ – LEVEL 8)

Project Handbook 2026-27

Ver 2627.05.20.b

Department of Computing & Mathematics

SETU WATERFORD

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Introduction

On this programme Work Placement is optional. **The Project is mandatory and must be completed successfully to graduate.**

As a final year student you are required to create a capstone project for your learning on this course. This will include software that you create along with additional artefacts to communicate what the project is, and your development and learning during the process. These artefacts include:

- I. Final proposal (2 pages max) with appended completed ethics checklist
- II. an **academic document** (final report) including sections for abstract, background, RAMP, AI (citing full chat/prompt/response history), reflection and critical evaluation of the project. It will include citations using Harvard Referencing, all relevant tables, figures, bibliography and appendices.
- III. a **single project repo with full git history** (github/gitlab) of your software development (including fully commented commits, branches, etc.). It may have links to sub repos. You must use **git** to manage revisions/commits (**daily**). **We will not attempt to evaluate your work without a full git history**. Include all artefacts in your repo so that you have a **documentation history** and backup also.
- IV. an implementation **plan** (in Jira/Trello or similar)
- V. a **video** demonstrating the software working < 10 mins
- VI. an **image**/poster/infographic summarising the project visually
- VII. a **presentation** which will be used in a live presentation at the end
- VIII. **showcase** details for the SETU Computing Expo
- IX. a hosted **project landing page** (webpage) with links to all artefacts created for the project.

In terms of size it should reflect the effort of a 15 credit module. (i.e. this is a substantial piece of work, way in excess of an assignment)

If you are unsure of how to cite or don't know what **Harvard Referencing** is, the SETU library provides guides, courses and hands on assistance. Please take the time to study the SETU library guide (SETU Libraries, 2017) before commencing your project. Tools help make the job easier – use the references tab in MS-Word to manage your citation in a word document. Alternatively, MyBib (2018), Zotero (2024), and Mendelay (2026) are good alternative free reference managers.

You are required to join the **#project** channel in **Slack** to receive all announcements including deadlines. It is your responsibility to check-in regularly for announcements/changes.

Before deep diving into a project, you must complete the **Complete the Ethics Checklist** (SETU, 2020) with the project coordinator. Key to this is “DO NOT use sensitive data (create a test data set instead)

We are experimenting and have produced a tutors site of projects for the 2024-2026 cohort which includes an LLM for each project for you to interact with (Tutors, 2026). Have a listen to the podcast if available, and definitely look at the infographics and mindmaps.

1. Project Proposals

The purpose of the project is:

- To provide a capstone for learning from the course
- Integrate knowledge from multiple modules
- Investigate new domains, technologies or procedures
- Document and record your work placement
- Showcase skills and achievements
- Formally research a new technical domain

You are invited to make project proposals on an individual basis. The proposal may reflect a deep personal interest, the interest of your supervisor, or aspects of the technology and business processes of your work placement. It may be useful to conceive a project as a solution to a problem. Representative project examples include:

i. **Work Based Project / Package**

A project allocated by your work placement supervisor, representing a reasonably independent piece of work you have been implementing on your placement:

- A standalone **application** (e.g. web app, mobile app, native or hybrid)
- A **component** of a larger system
- A new or improved **workflow / business process** serving some need in the workplace (e.g. CI/CD continuous integration / continuous deployment)

ii. **Independent Project**

Unrelated to work placement, consisting of an implemented software system designed to explore or consolidate knowledge in a specific domain

iii. **Research Paper**

Exploring a specific topic or set of topics and developed to conference paper standard. May or may not be related to work placement. This project type is **rare** as the course is focused on software development implementation.

The project may be directly aligned with your work placement opportunity, or it may be independent of it. This may depend on the disposition of the industry providing the placement, your personal goals and interests, and the advice of your project supervisor.

You will have **two meetings** in semester 4 to:

- discuss your project proposal,
- complete an ethical checklist,
- be advised on next steps.

If you are **working in tech**,

- i.e. placed by Placement Coordinator (Joan Mangan) otherwise you will need to submit your job spec to the Placement Coordinator for review
- You have the choice of a work based or independent project.
- We recommend **work based project** where you can make a part of what you are doing in work, your project (Killing 2 birds with 1 stone).

If you are **not working in tech**,

- You are incentivised to enhance your learning experience by undertaking **certification programme, otherwise your Project mark is capped at 70% but also floored at 40%** on successful completion of project. In other words, you drop a grade (10%) from the initial marks awarded after the project is presented, unless you undertake and complete certification training.
- You are, by definition, doing an **independent project**.

Free Certification Programmes are available to all students irrespective of whether you are working in tech or not, including **RedHat, AWS, IBM, Google AI, and Anthropic AI certification**. **Certification** can be undertaken by any student from September to August. Please take note of the deadline for certification completion (usually third week in August). We don't control or have the power to extend this deadline.

In **rare** circumstances the project may take the form of a research project – however this is generally not advised.

2. Project Timelines

Traditionally final year projects run over 2 semesters. You are completing a 30-credit module called ICT Placement Skills. It includes work placement, certification, and a project. Work placement is optional. **The project is mandatory and must be completed successfully to graduate.**

The HDip in Computer Science is an accelerated programme, and the project is accelerated also. However, to assist with balancing your workload, the project extends into a fifth semester and in some cases beyond.

The body of work constituting "The Project" will **commence in Semester 4**. Requirements, analysis, market & technical research, initial designs, modelling, and a sprint plan will be developed prior to your first supervisor meeting in Semester 5. "The Project" can be divided into 4 phases (Inception, RAMP, Development, and Dissemination)

Prerequisites for participating in Project

Ensure you have the follow pre-requisites in place:

- Computer (ideally laptop) to do the course.
- Camera (built in or add on)
- Microphone. (built in or add on)
- Slack desktop application installed and tested.
- ZOOM desktop application installed and tested.
- Security permissions for the mic and cameras, screen sharing etc should be granted on your system.
- Multiple web browsers installed (Install Chrome browser, Install Firefox browser)

i) INCEPTION Phase (June-November)

Includes formation of the project concept, two initial meetings (half hour each) with project coordinator. Draft and Final proposals + (optionally) interaction with workplace mentor, completion of ethics checklist.

Pre-Meeting 1 (June-September)

- Review past projects (SETU, 2023).
- Formulate project idea/proposal into a draft. Try answer the following:
 - What is it?
 - Why are you doing this? (what problem is it solving?)
 - What's the background?
 - Initial thoughts about how you are going to implement this. (What's involved)
 - If you are using data, consideration of the type of data.
 - If you think your project needs to be private - explain why and prepare an NDA.

Meeting 1 (October)

- You will submit a **draft** for each idea (max 2) in advance.
- The project idea(s) will be discussed with the project coordinator.
- Action from this first meeting is to select project and complete the Final Proposal within 2-3 weeks. Ideally get started on next phase

Meeting 2 (November)

- Arrive with final proposal and ethics checklist prefilled.
- Action from this second from this first meeting is to Review Final proposal (about 1-2 pages, 500 words approx.) and **Complete the Ethics Checklist** (SETU, 2020). Time permitting explore the scope the concept further. Ideally get started on next phase.

The Online Ethics Checklist

- This must be completed with your coordinator after the proposal is finalized. Review all materials related to the **Ethics Checklist** (SETU, 2020) in advance of meeting 2.
- **Q1 Final Year Undergraduate**
Q2 Short Commercial Name - Longer Academic Name
e.g. WufWuf - Progressive Web App for Dog Groomers
Q3 Colm Dunphy / TBC
Q26 <<Unique student code will be generated by the coordinator at meeting 2>>
- On completion with the project co-ordinator, you should save a copy as a pdf.
 - Right click, File | Print (Adjust settings to A4), Save As:
 - HDIP-Project<startyear>-<endyear>-<studentnumber>-EthicsChecklist.pdf
- When Submitting your final proposal, save your 2 page proposal as a pdf. Append the EthicsChecklist.pdf to it and submit (*usually before the first Friday in December*).
- It is inadvisable to proceed with your project without ethical approval.

ii) RAMP Phase – Research, Analysis, Modelling & Planning (November & December)

During this phase you will complete a lot of the documentation to help you hit the ground running in January. Create a document called **Final Report** and start populating with information gathered from the sections below. This document should be completed **before January** enabling you to jump into implementation in the new year. You will be asked for an **Interim Report** in mid Feb (Halfway point) which is a snapshot of this document with your reflection of your progress on the first three sprints.

Research & Analysis

With the proposal finalised, carry out:

- user requirements
- feature analysis
- market analysis (comparing similar products)
- repo analysis (repos on Github or Bitbucket) that address your idea. Maybe you can contribute to these projects or fork them?)
- technical research (frameworks, tools, APIs, etc.)
- Write up your findings
- Tables are good for summarising and concluding (e.g. use a table with a column for each alternative and a row for each feature). Make a conclusion based on the table.

Modelling / Design

This might be informed by creating personas for the users of the system. Then creating scenarios/walkthroughs of the system (telling a story about what happens next). In parallel you can create the actual models / diagrams:

- system model
- data model
- pipeline model
- wireframes for screens (consider using Miro, Figma, others)
- walkthrough with persona, scenarios

Use pencil and paper to mock up screens. Photograph them. Put each one on a slide. Use invisible buttons to jump to slide when clicked/tapped. This will give you an electronic paper prototype.

Now write the walkthrough story/stories referring to the labelled screens.

Update as necessary. Now look at the data that is required on each screen, this informs the data model / database. If not already done, complete during the Christmas period, ready to be reviewed by your appointed supervisor.

Methodology

Choose your Methodology - write a paragraph on your version of the chosen methodology - software engineering lifecycle/methodology e.g. (Agile, Sprint, Scrum, kanban, iterative, prototyping, test driven development, Waterfall, hybrid approach, etc.)

Plan

Make an implementation PLAN for 12 weeks (6 sprints)

e.g. use Trello/Jira to create cards for each task to be done. Estimate time for each on the card.

Calculate realistically how much time you are going to spend in each week working on your

project. Prioritise what features are needed for a minimal viable product (MVP). Maybe use User Stories for this. Estimate how long it will take for each user story (Manage all this with Trello/Jira or another alternative). Organise the cards/stories into stacks for each 2-week sprint (bearing in mind the estimated effort required and the time you have). Track your progress on the cards (RED for not done, Green Done, Amber in progress). Share with your supervisor in January. You have approximately 12 to be code complete (end of march).

Sample Plan Overview for Semester 5:

6 sprints of 2 weeks starting 1st Jan.

Sprint 1	Minimum Viable Product p1
Sprint 2	Minimum Viable Product p2
Sprint 3	Minimum Viable Product p3
Sprint 4	additional features
Sprint 5	additional features
Sprint 6	left empty for fixes/fancy features/ prepping artefacts
	dissemination

You may need to **rescope** your project - discuss with your supervisor in January.

If RAMP is not complete by the end of the year, you will eat into your implementation time and put yourself under unnecessary stress and pressure.

You must use **git** to manage revisions/commits (**daily**). We will not attempt to evaluate your work without a full GIT history of all your artefacts (code, reports, etc.).

iii) DEVELOPMENT Phase (January, February and March)

This is where you implement the project. You are responsible for completing the work agreed in your document/previous supervisor meeting. You meet your supervisor each week for half an hour – come prepared – updated Jira/Trello board. Communication should be through Slack as is the norm on this programme. Students should keep minutes and action points from each meeting (Attach to a card in Slack/Trello). This will help in the later stages of the project. At each meeting with your supervisor, user-stories/cards can be reviewed, and the next sprint (re)planned.

You must use **git** to manage revisions/commits (**daily**). We will not attempt to evaluate your work without a full GIT history of all your artefacts (code, reports, etc.) Supervisors should review progress in git regularly.

An **Interim Report** is due mid-February. After you submit, **seek feedback on this document from your supervisor** in preparation for submitting your final report. Ideally at the next meeting after submission.

A project landing page (web page) must be created and hosted (e.g. on Github or AWS.) Use Bitly (2026) to create a shorter URL to the landing page. Preferably edit the bitly shortcode to your commercial title (Perplexity AI, 2026). This short URL is a key requirement for the SETU Computing Expo. You can modify the web page later as long as you have the URL. Expect to republish this as the project proceeds. i.e. you update the links in the template as they are completed.

iv) DISSEMINATION Phase (April & May)

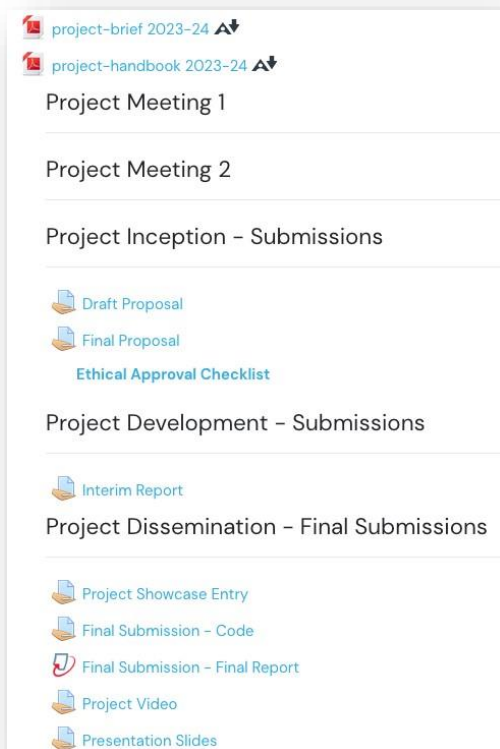
- I. Final proposal (2 pages max) with appended completed ethics checklist
- II. an **academic document** (final report) including sections for abstract, background, RAMP, AI (citing full chat/prompt/response history), reflection and critical evaluation of the project. It will include citations using Harvard Referencing, all relevant tables, figures, bibliography and appendices.
- III. a **single project repo with full git history** (github/gitlab) of your software development (including fully commented commits, branches, etc.). It may have links to sub repos. You must use **git** to manage revisions/commits (**daily**). We will not attempt to evaluate your work without a **full git history**. Include all artefacts in your repo so that you have a **documentation history** and backup also.
- IV. an implementation **plan** (in Jira/Trello or similar)
- V. a **video** demonstrating the software working < 10 mins
- VI. an **image**/poster/infographic summarising the project visually
- VII. a **presentation** which will be used in a live presentation at the end
- VIII. **showcase** details for the SETU Computing Expo. (project type, name, profile picture, project title(s), abstract, image representing the project, technologies/keywords, link to project landing page.
- IX. a hosted **project landing page** (webpage) with links to all artefacts created for the project (as listed above, may contain others).

Expect feedback in Slack and a requirement to resubmit quickly in the later stages of the project.

To help you plan your project, these **milestones** will serve as a useful guide:

Semester	Milestone	Description	When
4	1	Draft Proposal (Idea)	June-Sept Submit before Meeting 1
4	2	Meeting 1 with Project Coordinator	October
4	3	Draft Proposal Updated	Before Meeting 2.
4	4	Meeting 2 with Project Coordinator	November
4	5	Final Proposal	Ideally before Meeting 2. latest Early December
4	6	Online Ethical Checklist https://moodle.wit.ie/course/view.php?id=176894 <i>see Appendix 7</i>	Completed at Meeting 2
		RAMP Phase Research, Analysis & Design, Methodology, Plan	Oct, Nov Dec.
		Individual Project Supervisor Appointed	end of semester 4 / start of semester 5
5	7	Interim Report	Mid-February (Sunday after Valentines Day)
5	8	Showcase Entry (including image/poster/infographic, landing page, profile picture, abstract, titles)	Week 10
5	9	Final Project Submission (code-complete)	Week 12
5	10	Final Project Submission (final report) submitted through Turn it In / Tutors Agent	Code complete +1 day
5	11	Video demo of project	Code complete +1 week
5	12	Presentation slides	Code complete +1 week
5	13	Presentation of Final Project to judging panel	Code complete +2 weeks
5	14	Initial Grade of Final Project	Just Before Expo
5	15	Presentation of Final Project at http://expo.setu.ie	Before Week 4 in May

3. Project Submissions



Introduction

During project development, several artefacts are created. Before commencing, please familiarise yourself with the requirements of this programme in relation to AI and git, and SETU's policies (SETU Academic Council, 2026) in particular:

Policies

- Student Expect Behaviour Policy
- Student Academic Integrity Policy
- Use of Gen AI Policy.

AI

As computing students, we encourage the transparent use of AI tools. They must be acknowledged, cited and used appropriately. For this programme, we require an AI chapter in your report listing the AI tools used on the project, an evaluation of each tool, evaluation of AI use overall on the project. Provide citations for each tool which enable us to see the full chat history used on your project (prompt, responses, and timestamps).

git

A full GIT History is required in your repo for the duration of the project, for your code, documentation and artefacts.

Proposal

The draft proposal will articulate the project concept and nature and will serve to scope the work of realising the project. It may typically be less than **<2 pages**. For work-based projects you need to identify and get permission for your project from the employer, Flag NDA requirements now. NDA Scope should be restricted. (see NDA Policy below). The final proposal submission will have your completed ethical approval checklist appended.

Ethical Approval Checklist

To ensure your project is viable without requiring formal ethical approval from the university's ethics board (with a 3-month delay), you will need to complete this checklist. This may involve modifying the scope or nature of your project. Without successful completion you cannot submit the project. This will be completed during Project Meeting 2.

Final Report - RAMP

This document should be completed before January and submitted to your project supervisor in advance of your first meeting. It will introduce the project, give the background, show your RAMP.

Interim Report

This will be a substantial update on the progress of the project – and can be regarded as an early draft of the final report. It may be **up to 4000 words**, depending on the project type. This report is not marked but you should seek feedback on it from your supervisor to assist in completing the project and your final report. It should contain RAMP plus the first half of your reflection. The interim report will be present and future tense – where are you now and what will be done in the remaining weeks.

Include your calculations for wordcount

e.g. Document words = 5000 – appendices 1500, TOC 300, Bibliography 900, Headings 300 = 2000

Showcase Summary

A showcase of all Computing projects is produced each year for the SETU Computing Expo. Look at last year's brochure, to better understand its purpose. To automate its compilation, we will require several small details which are requested through an online form or markdown file. Character limits will be enforced on each field. We will use this data to populate an initial Tutors website from which you will need to do pull requests to make changes, so it's best to get it right from the start. What's requested:

- Student number
- Name
- Personal Digital Portrait Photo (headshot from this year with plain background)
- Project image/infographic (input/process/outputs. Tools used. screenshots)
- Commercial Title (40-character short name for project)
- Academic Title (100-character explanation of what it is)
- Abstract / Summary (500 to 600-character informal description of project)
- Technology Keywords (max 100 characters. Comma separated list of technologies used.)
- Supervisor's Name (Title First name Lastname)
- Project Areas (select all that apply)
- Link to Project Landing Page
- NDA?

Additional Specifications / assistance:

- Personal digital **portrait photograph** - use a clear up to date (tall rectangle) with a plain contrasting background. Width 600 * Height 800 pixels, saved as a .PNG or .JPEG file. Use <https://pixlr.com/> to edit it using the Crop tool. You should be instantly recognisable from this photo.
- **Project image** should be an A4 portrait image Width 2480 * Height 3508 pixels saved as a .PNG or .JPEG file
- **Commercial title** (project name) e.g. "FindThatBook.com"
- **Academic title** of the project e.g. "distributed library book retrieval system".
- **Project Landing Page URL** - Project webpage (shorten the link using **bitly**). This is a web page which acts as a home/landing page for your project. You can add links to this page as the progress continues. Use github pages, S3, netlify, your own web hosting. Create a single web page, with name, project, photo, and additional links for the project which will include all the artefacts requested from your project plus any others that you wish to add.
- **Project Type** – Select all that apply e.g. (AI/ML Development; CI/CD, Testing, DevOps; Cloud Computing; Computer Networks; Database and Analytics; DevOps; IOT/Physical Computing; Open Source; Personal/Independent Project; Software Development: Back End, Core, Front End, Mobile – Hybrid; Mobile – Native; Web; Work Based Project)
- **Technologies/Keywords** e.g. (React, Flutter, Kubernetes, AWS).

Final Code

Your codebase is a single repo with full git history (it may contain links to sub repos). Provide the public link to GitHub, GitLab, Bitbucket, etc. If it needs to be private due to being a work project, send invite to the project panel.

Video

The **video** must be hosted (e.g. YouTube). Max 10 mins without exception. Use your student account on YouTube. Make the video unlisted or public. Both the link to the video on YouTube and the original video will be submitted in Moodle. See Appendix 2 for more details on the video.

Presentation

When presenting the final project in person, you will prepare a presentation e.g. Powerpoint. This **presentation** will be submitted just before or straight after your presentation.

Final Report

The final report is **up to 8000 words**. The final report is written in the past tense – what you did. Remember images don't affect your word count. Your final report should have most of the following features:

- a cover page
- declaration of authenticity
- acknowledgements
- titles (commercial - academic)
- preface
- abstract
- keywords
- glossary
- list of abbreviations
- TOC, TOF, TOT, appendices, and a bibliography.

The TOC will vary depending on the project (write to the assessment criteria, and the development of the project). Make sure to include RAMP, implementation details, AI section, Reflection (what you achieved, learned, future development, problems encountered and solutions).

Use numbered headings and styles, page numbers, **and Harvard Referencing**.

Use numbered captions on tables and figures.

After the final report deadline, the upload facility will be closed (cut-off) and the reports downloaded and distributed to the panel for review.

You must cite and reference properly. We are looking for your words not an AI generator's words (which must be cited). This is particularly important in the Reflection section.

Arrange to submit a draft to your supervisor allowing time to receive constructive feedback to be acted on before final submission.

Include your calculations for wordcount

e.g. Document words = 10000 – appendices 1500, TOC 300, Bibliography 900, Headings 300 = 7000

Artefact Usage

Artefacts produced and submitted as part of the project (including reports, images, video, code, and slides) may be used for future quality, training, and promotion purposes. Final projects may be archived, as part of the Institute's efforts to provide a repository of the interesting and valuable work carried out by its members and to help future students. Sensitive work can be excluded by way of NDA subject to NDA policy. Confidential Appendices should be separate documents and password protected.

If your project is shortlisted for the Best Project Award you will be contacted to decide if you want to give permission for the project artefacts to be reviewed by the award sponsor.

4. Submitting

You will be required to adhere to the milestones outlined in Section 2. All submissions are uploaded / linked from Moodle. Soft copies are required. Hard copies are no longer required.

NDA Policy (subject to amendments)

In encouraging students to take on work-based projects, we understand that some projects may be sensitive. We have used NDAs in the past to facilitate these. However, blanket NDAs for all student work is frowned upon. Where a project is sensitive to a particular company for work / work placement, **if NDAs need to be signed these should be submitted at proposal stage.** The scope of the NDA should be constrained and needs to be discussed early on.

Based on past experience NDAs are requested for five reasons (1. Sensitive data; 2. Sensitive code; 3. Sensitive manager/company/industry; 4. So a student doesn't have to share their work; 5. Because we can!)

Some guidelines:

1. **Student projects can no longer work on live personal / sensitive DATA** – as they fall foul of ethical approval. (They can use anonymised test data for real world systems being developed.) Students will need to use and or produce this test data. AI is ideal for this. NDA not applicable.
2. **The project review panel must be able to review and mark CODE produced by the student.** If this is not possible, the work is unsuitable for a university project. Code does not have to be made public. Company code libraries used in the project do not need to be shared/submitted. A private repo is preferred with limited time access to the project panel of three. We will consider alternative arrangements where this is not feasible.
3. We appreciate that software being produced in **some industries may be more sensitive** than others. We will review on a case by case basis.
4. **We won't NDA work just to make it private for a student.**
5. **NDAs will be signed if there is good reason, not just because it is a project for a company.**
6. **Students have to produce artefacts as part of their project.**
 - Blanket NDAs will prevent any mention of anything to do with the project in Computing Expo etc. We would prefer students to be able to:
 - Describe their project (without necessarily mentioning the company) for their Showcase Abstract
 - Use a more abstract title for their project if this helps
 - Produce and share a graphic about the project (possibly more abstract if this helps)
 - Produce a video demo of the project (which can obscure company name etc., for example)

- Produce a more abstract final report which can refer to separate private appendices.

When processing the Showcase Handbook entries, we would like to share as many artifacts as possible for each project. We would encourage each company/student to discuss early on, what can be shared and how this is reflected in the individual NDA.

Students still need to submit ALL the deliverables as normal plus additional data as requested, after NDAs are submitted. If covered under NDA they should be flagged. We will respect NDAs and we will keep the details of NDA projects private where NDAs are suitable and have been submitted and signed in advance.

5. Declaration of Authenticity

Immediately inside the cover page, include a signed declaration shown in Appendix 5.

6. Word Count

4000 words for Interim Report

8000 words for Final Report

You don't get to write all that you want. You have a limit. Break up each section and give it a budget wordcount. Work towards hitting the count for each section. Fill in each section without much thought of wordcount. Then rewrite to meet it. Your writing improves. Your communication improves. Waffle disappears.

If you are finished and you are over then consider that the doc is your draft 1.

Show the calculations for Wordcount I Appendix 1 so that so that it doesn't affect the wordcount. e.g.

Document word count: 11,300

Less:

TOC, TOT, TOF: 500

BIBLIOGRAPHY: 1000

captions: X

headings: X

TOTAL: 9800

This might help you get the count down. You will learn more working on this manually.

Look at what's in your report. What isn't essential to read that's taking up word count? >> Move to an appendix. Usually, you can get to where you need with this approach.

If you have thousands of words over. Then you need help with editing and rewriting. If you must, make sure you have the original document backed up, then prompt AI to act as an editor, to reduce the wordcount [make sure you save the version before it edits!].

Finally within but not over 10% of wordcount is generally acceptable [but you always check with the reader]. In this case we would accept within 10%.

7. Assessment

In preparing your project for assessment, you should consider both core criteria and a critical self review: The project is marked as a whole, without a section breakdown. However, for guidance consider

PART A - Core Criteria (85%)

Level (25%)

Appropriate mix of (a) **originality**, (b) **innovation** and (c) **complexity**.

Communication (30%)

- Documentation:
 - Submitting all **artefacts** as requested and on time
 - Git history
 - AI Section
 - Reflection
 - Writing well-constructed formal reports (tables, numbering, Harvard Referencing & citations)
 - Trello board (used for plans and progress).
 - Sketches of ideas in diagrammatic/written form, wireframes
 - Project Image
 - Video demo
 - Live Presentation
- Modelling:
 - lightweight, relevant modelling, generally in accordance with a recognised process. Use Appropriate diagrams including UML, Architectural Model, Process Flow Charts, Data Models (E-R), UI wireframes etc.
- Communication with Supervisor & Coordinator throughout including Slack, meetings.

Implementation (30%)

Based on the modelling and the content of the reports. (i.e., Have you implemented what you said, or something different?) How much? Working?

PART B - Critical Self-Review (15%)

- what you **learned (5%)**
- what you **achieved (5%)**, and in what direction the project might be taken if more time was available (**future development**)
- **problems** encountered (**5%**) how they were addressed and solved

Note:

- Penalties will be applied for **late** submissions, **exceeding** the character count, word count, or maximum durations specified. Penalties will also be applied for not submitting what is requested, in the proper **format**, or for **not citing** or acknowledging (plagiarism).
- All students must include an **AI Chapter** listing the AI tools you used on the project, an evaluation of each tool and AI use overall on the project. Provide **citations** for each tool which enable us to see the **full chat history** [prompt and responses].
- Projects can no longer be evaluated without a **full GIT History**.

The Assessment Process

Marking is done at the end of the project. The Project Grade Categories considered are (outstanding 85+, excellent 70-84, v.good 60-69, good 50-59, fair 40-49, fail <40).

Before Live presentations begin:

- A schedule will be created grouping the projects by type (so that similar projects are scheduled and assessed together).
- Presentations will be scheduled, maximum six per day.
- A panel of 3+ lecturers will be tasked with reviewing all projects. The panel makeup will be communicated to all in the week before presentations. For large classes, additional panels may be created, and each panel will look at a subset of all projects.
- Your supervisor will provide marks to the panel. These are for indicative purposes only.
- All panel members will have viewed your demo video.
- At least 1 member will have read your report in detail.
- At least 1 member may have reviewed your code repos.

At the Live Presentation

- All panel members will be present for your presentation. Your supervisor may attend also. A subject area expert may also be invited.
- You have a max of 15 minutes to present your project followed by Q&A.
- After each session, the panel member will have an indicative ranking relative to each project on that day. At the end of each day the projects for that day are discussed among the panel in a collegiate fashion and ranked initially starting with highest and lowest working towards the middle. Previous days will be revisited and re-ranked at the end of each day/week.
- When all project categories have been completed, the panel will break and meet again on a separate day, the panel will review and discuss the projects again attempting to merge the different categories.
- Then, marks will be applied based on the Project Grade Categories.
- Borderline cases will be discussed again.
- If there is an unacceptable difference between a supervisor's suggested mark and the panels allocated mark, the project and mark can be reviewed again.
- Where there are more than 1 panel, there will be an additional step of reviewing and merging the results of each panel.

The External Examiner will review the marks and may suggest changes.

8. Project Supervision

Each student will be appointed an academic supervisor to provide supervision and advice. The norm will be that you will have a coordinator for 2 meetings in semester 4, and a supervisor appointed for semester 5. It is recommended you discuss the project progress at regular intervals with your supervisor, usually face to face (e.g. Slack/Zoom call). Submit deliverables to your supervisor first a week in advance to allow time for constructive feedback.

Project Meetings

There will be 2 * 30 minute meetings in the inception phase with the project coordinator.

Most meetings will occur in semester 5 during the development and dissemination phases. Usually, each project student will have a weekly meeting with their supervisor consisting of (12 half hour meetings or 6 one-hour meetings) at the discretion of the supervisor. Students will show his/her progress each week against their plan.

For each meeting, brief **minutes** and **action points** should be taken by the student. Consider creating cards and updating progress using Trello or similar (make sure to give you supervisor access). Keep a summary each week outlining what was agreed and what is to be done for the next meeting. Follow-on meetings can start by reviewing these minutes and action points.

If either the supervisor or the student is unable to keep a project-related appointment, the other party will be informed as soon as possible using **#Slack**. Naturally it will be rare for such cancellation to take place and an alternative should ideally be arranged.

9. Project Finishing Date

The module will have an oral examination/demonstration in April/May. It is crucial to get started early and work on it each week, particularly in the implementation and dissemination stages.

Projects will be completed at this stage

Repeats will submit in August.

For capped independent projects where students want to complete certification over the summer, the project placement module is deemed to be complete in August. Initial Project marks may be updated to take account of certification for those not working in technology.

Project Marks will not officially be released until September irrespective of when you complete.

10. FINAL MARK - How does Work Placement, Project, and Certification come together?

As mentioned previously, the module title is called “ICT Skills Placement”. It incorporates Work Placement, Project, and Certification.

The **Placement Co-ordinator** (Joan Mangan) will request a **job spec** for your work placement or existing ICT role if upskilling, to advise if suitable or not. A **Moodle Dropbox – Job Spec/ICT Role** will be provided for this.

Towards the end of placement, The Placement Co-ordinator will review your experience by first requesting your learning log (which might be recorded through a blog / vlog / competency document, etc.). This will be used to review your experience in preparation for a site visit and interview with your mentor. A result of pass / fail is recorded

If the spec is suitable and you completed sufficiently, this is recorded positively. If the spec is unsuitable this is recorded negatively and you are recommended to complete the **certification** to maintain your indicative project grade. Proof of completion will be submitted to a **Moodle Dropbox – Industry Certs.**

Indicative project grades are given for the Project after correction in April/May. You will have the option to complete certification which will be checked at the end of August.

Adjustments:

Placement / existing ICT	Certification completed successfully	Initial indicative project grade	Module Mark
Y	Y	>= PASS	Remains same
Y	N	>= PASS	Remains same
N	Y	>= PASS	Remains same
N	N	>= PASS	Reduced by a grade (10%). (Cap of 70, floor of 40)
		FAIL (in April)	Review again in August
		FAIL (in August)	FAIL MODULE, don't graduate. MUST REGISTER TO REPEAT MODULE FOLLOWING YEAR (CURRENT COST PAYABLE BY STUDENT €1200 – reduced from €1675)

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Appendices

APPENDIX 1: Project Proposal Outline

- Your project proposal must have a **title** (see below).
- Project Type (Word Based / Independent / Research)
- Project Category (Web App, Native Android, Hybrid, CI/CD PIPELINE, RESEARCH, Other)
- Your project proposal will be about **four or five hundred words long**.
- Please re-read your proposal when written, and amend it if necessary, to ensure that it is easy to read and makes sense to another person.
- You may wish to include a diagram or two.
- You should provide a list (with a title) of **technologies** that your project might employ, both software (including programming languages) and hardware.
- You should provide a list (with a title) of **tools** and frameworks that you intend to use.
- You should indicate the project **process** you intend to use: for example: whether you intend to implement your software all at once or as a series of production quality releases where more functionality is added as you go along.
- You should indicate other parties with whom you are collaborating or parties for whom the software is being written. These are sometimes collectively referred to as **stakeholders**.
- A title page is important to ensure that we can easily file and retrieve your report. Please provide a page at the front carrying the following information:
- At the top of document include:
 - Your programme: Higher Diploma in Science in Computer Science.
 - Final Project Proposal
 - A title of your proposed project
 - Your name and student number

APPENDIX 2: Video Guide

This is a simple video **ideally 3-7 mins** in duration, but we will allow up to a maximum of 10 minutes (The panel are instructed to stop playing at 10 minutes). It is a demonstration of the key features of your project. It acts as a backup in case you can't get the live demo working on presentation day. It also provides a convenient way to see the project working. It helps focus the mind on the creating a good presentation.

It is not expected to be a Hollywood production. Depending on the project you might:

- use a screen recording to do a walkthrough of the software implemented
- use a phone (in landscape mode) to record a video of some hardware you are controlling with software.
- Do a video recording highlighting the key elements of a research project

The video must have **voice commentary (your voice)** explaining what's happening. It must use your voice explaining what is happening. Typically, I would suggest start by introducing yourself "Hi my name is X" and my final project is called Y. One sentence description of the system. Then demo.

Most systems have registrations & logins, we take it for granted that this functionality is working at this stage. By all means, login/register if you wish (quickly) but don't labour the point and don't waste too much time in demonstrating how great your login/registration is. We are more interested in **what makes your system different. Focus on the key aspects of your project.**

If recording mobile, ideally check if your software has a trails feature, which makes it much easier to see the user interaction.

Video Software Recommendations:

Desktop:

You've already used **OBS** to record videos, so this is a fine way of recording the desktop.

Camtasia is the best screen recorder/editor on PC.

ScreenFlow is the best screen recorder/editor on Mac.

Demos are available (don't worry about watermarks on any software as long as they are not excessive)

Jing is free.

Loom is free.

There are others.

Mobile:

Built in screen recorders.

Camtasia & ScreenFlow record from your mobile device too.

Zoom

Android:

DU Recorder Screen Recorder

iOS:

RecordIt!::Screen Capture

Submitting

There are **two parts** to submitting:

URL

Upload your video to YouTube or similar (make it unlisted or public). Copy the URL. Paste it in the 'Online text' textbox. Highlight it and make it a hyperlink by clicking the link button and pasting the URL again.

FILE

Upload your video file to Moodle also in the File submission box - you can just drag and drop. It will most like be an mpeg4 encoded video (.mp4)

Project Video

<https://tutors-design.netlify.com/talk/wit-hdip-comp-sci-2018-sept-onsite.netlify.com/topic-01-project/talk-5-project-handbook/project-handbook.pdf>

File submissions

Maximum file size: 2GB, maximum number of files: 3

Files

ZoomBreak...

Online text

<https://youtu.be/w8Uwy-9YXkY>

Save changes Cancel

When you click Save, your Moodle submission will look something like this.

Project Video

<https://tutors-design.netlify.com/talk/wit-hdip-comp-sci-2018-sept-onsite.netlify.com/topic-01-project/talk-5-project-handbook/project-handbook.pdf>

Submission status

Attempt number	This is attempt 1.
Submission status	Submitted for grading
Grading status	Not graded
Due date	Sunday, 31 May 2020, 11:55 PM
Time remaining	2 days 11 hours
Last modified	Friday, 29 May 2020, 12:11 PM
File submissions	 ZoomBreakoutRoomGuide.mp4 + 29 May 2020, 12:11 PM Export to portfolio

Submission comments [▶ Comments \(0\)](#)

Online text



[Export to portfolio](#)

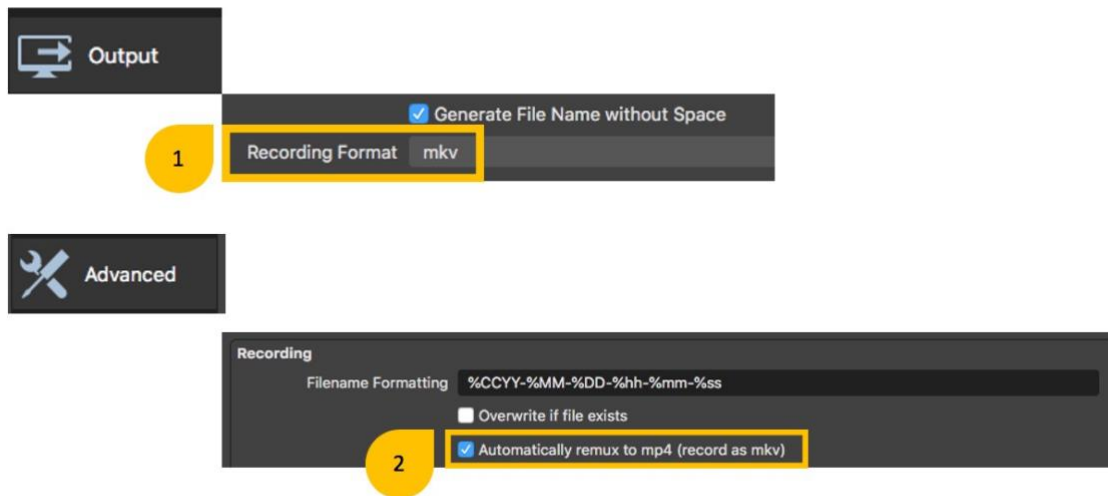
[Edit submission](#)

[Remove submission](#)

You can still make changes to your submission.

Additional:

Tip for recording in OBS



.mkv & **.mp4** about 10 seconds after you press stop

APPENDIX 3: Presentation Guide

The week before your presentation, create your slides. These will be submitted to Moodle. It's a good idea to arrange & rehearse presentations a few days before with each other. Honest critiquing will help everyone improve.

You have up to **15 minutes to present**, followed by up to **25 minutes of Q&A**. So, this time will need to be managed well.

Planning your Presentations:

Part 1 – Introduction

1. Introduce yourself.
2. Give the name of the project and your short abstract (1 or sentences)

Part 2 - Sales Pitch (everything is great)

1. Demo what you've got
(This is ideally a live demo, but just in case it's good to have a recorded backup)

It's a good idea to use **real data** to help with understanding the system (i.e. don't use "test")

Part 3 - Academic Explanation (SUMMARY OF KEY POINTS OF YOUR REPORT)

1. i.e. how did you built it from start to finish? Analysis, design, plans, methodology, artefacts created along the way.
2. Summarise sections of your final report. What you learned. Key milestones. (demo videos). Problems encountered & solutions. Future development

Part 4. Wrap up

Any Questions + ask if anyone wants to try it out (live demo again)

Typical Timing

Part 1 & 2 will take 5 mins.

Part 3 & 4 will take 10 mins.

Q&A will take up to 25 minutes

Tips

- Split up what you want to say and allocate a time budget to achieve it.
- Summarize and Visualize
- Use images instead of text where possible
- Short videos can also help
- A good yardstick is have no more than 7 points on the screen at any given time
- Max of 25 words on screen (there are exceptions)

JUNE 2020

“RUNNING SOCIAL”
Cloud Native App Development Pipeline
FINAL REPORT

NAME:

STUDENT ID:

LECTURER: COLM DUNPHY

MODULE: PROJECT

COURSE: Higher Diploma in Computer Science

Table of Contents – Samples

Visit the past project archive at SETU Computing Expo (SETU, 2023) and review the final reports available from the project landing pages.

The Table of Contents for older projects will be missing the AI chapter.

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✓ Background

-  Current Features

✓ Objectives

-  Main Focus of the Project
-  Minor goals of the Project

✓ Methodology

-  Project Management
-  Programming Languages
-  Deployment Technologies
-  3rd Party Solutions/Technologies

✓ Project Plan

-  Phase 1 Inception
-  Phase 2 Development
-  Phase 3 Dissemination

✓ Development

-  Proof of Concept
-  Decouple UI from back end Server
- >  Building Container Images and Setup of Docker-compose
- >  Evaluating OpenFaas for Serverless functions
-  Refactor back end server to serverless functions
-  Final Testing and Deployment

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-  Minor objectives of the project

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APPENDIX 5: Declaration

I declare that the work which follows is my own, and that any quotations from any sources (e.g. books, journals, the internet) are clearly identified as such by the use of 'single quotation marks', for shorter excerpt and identified italics for longer quotations. All quotations and paraphrases are accompanied by (author, date) in the text and a fuller citation is the bibliography. I have not submitted the work represented in this report in any other course of study leading to an academic award.

Student..... Date

Work Place Mentor..... Date

APPENDIX 6: Academic Writing

If you are unsure of how to cite or don't know what **Harvard Referencing** is, the SETU library provides guides, courses and hands on assistance. Please take the time to study the SETU library guide (SETU Libraries, 2017) before commencing your project. Tools help make the job easier – use the references tab in MS-Word to manage your citation in a word document. Alternatively, MyBib (2018), Zotero (2024), and Mendelay (2026) are good alternative free reference managers.

APPENDIX 7: Sample Ethics Checklist for School of Science & Computing

- **Q1 Final Year Undergraduate**
Q2 Title = Short Commercial Name - Longer Academic Name
e.g. WufWuf - Progressive Web App for Dog Groomers
Q3 Colm Dunphy / TBC
Q26 <<Unique student code will be generated by the coordinator at meeting 2>>
- Pay particular attention to Q16 and Q22.
- Training materials for Ethics Checklist are provided in Moodle

 [News](#) [My Home](#) [Calendar](#)  12  

 Students/applicants must **answer all questions** presented to them until the Checklist questionnaire is completed.

1

Are you a student on a **final year undergraduate** programme, a **taught postgraduate** programme, or are you conducting a **research project**?

☒ Final Year Undergraduate
☐ Taught Postgraduate
☐ Postgraduate Research Project
☐ Other Research Project

2

What is the **working title** of your project?

AlexaRIP – an alexa skill integrated with RIP:IE to provide a voice solution to the daily dilemma of who died

3

Who are the project **supervisors/advisors/principal investigators**?

Colm Dunphy / TBC

4

Does your project involve **human experimentation**?

☐ Yes ☒ No

5

Does your project involve **live animal experimentation**?

☐ Yes ☒ No

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Ethics Checklist for Undergraduate, Taught Postgraduate and Research Projects in the School of Science and Computing (2025–2026)

[Questionnaire](#) [Preview](#) [Resume questionnaire](#) [View all responses](#) [Non-respondents](#) [More](#) ▾

This Ethics Checklist must be completed for all final year undergraduate, taught postgraduate and research projects in the School of Science and Computing.

7



Does your project involve the use of **human** remains/cadavers/tissues/cells/biological fluids/embryos/foetuses?

☐ Yes ☐ No

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9



Does your project involve the use of **animal cells, tissues or biological fluids**?

☐ Yes ☐ No

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11 Does your project involve the **collection of rare or protected plants**?

 ☐ Yes ☒ No

12 Does your project involve the generation or use of **genetically modified (GM) plant material**?

 ☐ Yes ☒ No

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14 Does your project involve the use of **ionising radiation**? (e.g. use of gamma ray spectrometry)

 ☐ Yes ☒ No

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16

Does your project involve the **collection of any new (or primary) data** from **individual people or groups**?



☐ Yes ☒ No

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21

Does your project involve **persons under the age of 18 years** (i.e. minors), or **any vulnerable groups**? (e.g. prisoners, refugees, those in care, addiction service users, etc.)



☐ Yes ☒ No

22

Does your project involve the use of **existing (or secondary) human data**? (i.e. data originally collected for another purpose)



☐ Yes ☒ No

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This Ethics Checklist must be completed for all final year undergraduate, taught postgraduate and research projects in the School of Science and Computing.

25 Before you submit the Ethics Checklist, you must **confirm all of the following**:



- ☒ I understand that the Ethics Checklist is a formal declaration.
- ☒ I have answered all questions on the Ethics Checklist carefully and truthfully.
- ☒
The supervisor/advisor (or principal investigator) for the project is present as the Ethics Checklist is being submitted, or they have given me explicit permission to submit it in their absence.
- ☒ I have had adequate ethics training and/or instruction prior to completing the Ethics Checklist.
- ☒ I understand, and agree to abide by, the general ethical principle of "do no harm" for this project.
- ☒ I will follow the instructions given in the Feedback Report.

26 Authentication Code (ask your project supervisor/advisor for this code)



Enter Student Number:

Enter the Authentication Code below and click "Verify Code"

Note: If an INVALID authentication code is used then this submission is NULL and VOID

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[Submit Ethics Checklist](#)